

Abstract

The aim of the project is to understand and utilize the concepts of Natural Language Processing in classifying the human and machine generated multilingual texts. The implementation mainly focuses on extracting particular language data - English, Spanish, Czech and German, which is processed by creating the word embeddings and vectorizing based on the model requirements. Later fed to the models - SVM, BiLSTM, mBERT and XLM-R. The data used is a combination of two datasets MULTITuDE and MultiSocial, in which the required language data is extracted and balanced human and machine generated texts summing to 26,671 samples which are used to train and test the models. Further, the trained models are tested on other languages - Portuguese, Dutch and Russian to analyze the cross-lingual transfer learning for human and machine generated text classification. Results are analyzed based on accuracy and F1-Scores.

Goal of the Project

- Use the above mentioned models in classifying the human and machine generated texts.
- Process the data to balance, having equal samples for human and machine generated texts with identified languages.
- Analyze and compare the performance of mentioned pre-trained models over human machine text classification for multilingual data.
- Perform cross-lingual test of the trained model and analyze the results.

Dataset

- The dataset is a combination of two datasets - MULTITuDE v3 and MultiSocial.
- Languages used for the training English, Spanish, Czech and German.
- Total of 26,671 samples of data used with each human and machine texts accountin g to 13,336 samples of texts.
- Cross-lingual data subset extracted from the dataset combination with languages used - Portuguese, Russian and Dutch with a sample size of 7,796 texts balanced between human and machine generated.

Methodology

- Data Preparation:** Combined MULTITuDE v3 (news domain) and MultiSocial (social media domain) datasets based on the length, filtered for 4 target languages, and balanced to achieve 50/50 human-machine distribution with 26,671 total samples.
- Data Split:** 80% training (21,336), 10% validation (2,667), 10% test (2,668). Stratified by language and label to maintain balance.
- Method 1 - SVM:** Traditional machine learning using TF-IDF features (1-3 grams, 10k max features) with Linear SVM classifier. Represents text as sparse bag-of-words vectors.
- Method 2 - BiLSTM:** Deep learning approach with custom word embeddings (vocab size: 1,24,807, embedding dim: 128). Bidirectional LSTM (2 layers, hidden dim: 128) captures sequential patterns and context from both directions.
- Method 3 - XLM-R and mBERT:** State-of-the-art transformer model pre-trained on 100 languages. Fine-tuned for 5 epochs with learning rate 0.00001 which changes continuously, batch size 4, using pre-trained multilingual embeddings.
- Evaluation:** Tested on seen languages (en, de, cs, es) and unseen languages (pt, nl, ru) to measure cross-lingual transfer.

Results

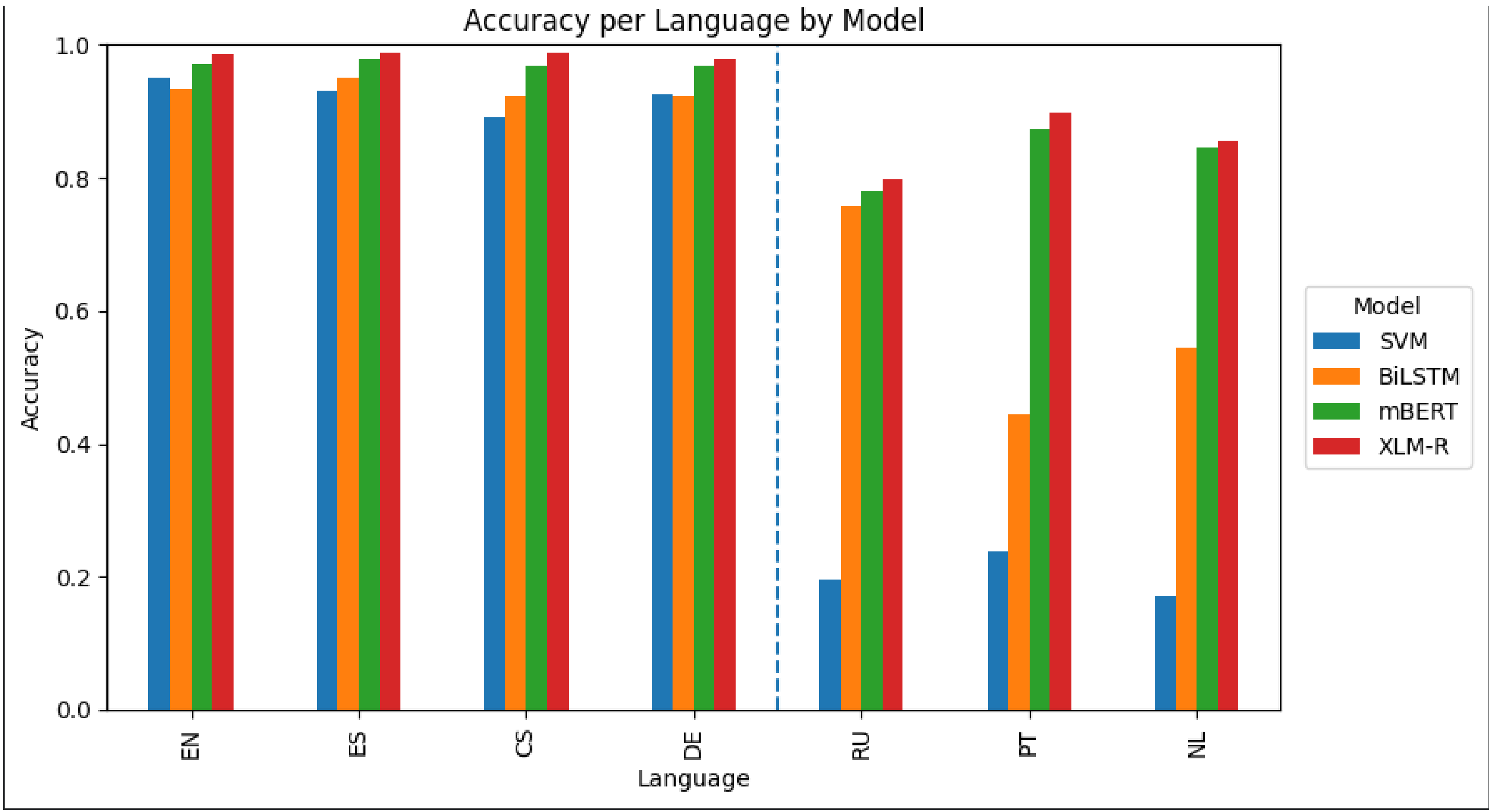


Image 1:Image showing the accuracies of different models per language.

Language	SVM	BiLSTM	mBERT	XLM-R
Trained languages				
English	0.9505	0.9340	0.9700	0.9865
Spanish	0.9310	0.9505	0.9791	0.9888
Czech	0.8904	0.9234	0.9694	0.9877
German	0.9251	0.9237	0.9684	0.9779
Overall –	0.9243	0.9329	0.9719	0.9854
Unseen languages				
Russian	0.1962	0.7584	0.7793	0.7984
Portuguese	0.2372	0.4447	0.8732	0.8986
Dutch	0.1706	0.5434	0.8447	0.8559

Table 1:Accuracies of models per language

Analysis

- From the above results it is clear that transformers are better approach in the task to classify the multi-lingual human and machine generated text. Out of thwo BERT varients trained, XLM-R has shown better results as it is pre-trained on a large set of data.
- Although the BiLSTM and SVM were able to perform the required task but they were not as robust and generic as the BERT models as per the results, as they are trained on very small vocabulary and due to simple TF-IDF method.

Limitations and Future Work

- Implementation of the process followed on larger and complex dataset with high multi-lingual samples and analyzing the results over multiple languages and as a whole.
- Exploring other variants of BERT models and other efficient models that are used for classification.

References

[1] D. Macko, J. Kopal, R. Moro, and I. Srba, “MultiSocial: Multilingual benchmark of machine-generated text detection of social-media texts,” *arXiv preprint arXiv:2406.12549*, 2024. doi:10.48550/arXiv.2406.12549.

[2] Y. Wang, J. Mansurov, P. Ivanov, J. Su, A. Shelmanov, A. Tsvigun, O. M. Afzal, T. Mahmoud, G. Puccetti, T. Arnold, C. Whitehouse, A. F. Aji, N. Habash, I. Gurevych, and P. Nakov, “SemEval-2024 Task 8: Multidomain, multimodel and multilingual machine-generated text detection,” *arXiv preprint arXiv:2404.14183*, 2024. doi:10.48550/arXiv.2404.14183.